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Diagnosis information - Gateway

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Diagnosis information - Gateway control module

Current distributor, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

Diagnostic information - gateway control unit

Power distributor, terminal 30F relay

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs. In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Function restriction

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 Mohm**

Open circuit :

- ♦ Setpoint resistance **< 2 ohm**

Short circuit between lines:

- ♦ Setpoint resistance **> 0.9 Mohm**

Diagnostic information - gateway control unit

Power distributor, terminal 30SD relay

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs. In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Function restriction

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage - **0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance > **0.9 Mohm**

Open circuit :

- ♦ Setpoint resistance < **2 ohm**

Short circuit between lines:

- ♦ Setpoint resistance > **0.9 Mohm**

Diagnostic information - gateway control unit

Power distributor, terminal 15 relay

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs. In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Function restriction

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 Mohm**

Open circuit :

- ♦ Setpoint resistance **< 2 ohm**

Short circuit between lines:

- ♦ Setpoint resistance **> 0.9 Mohm**

Diagnosis information - Gateway control module

Current distributor, Pyrofuse

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Open circuit

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Diagnostic information - gateway control unit

Current distributor, transport connector implausible

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs. In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Incorrectly set

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage - **0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance > **0.9 Mohm**

Open circuit :

- ♦ Setpoint resistance < **2 ohm**

Short circuit between lines:

- ♦ Setpoint resistance > **0.9 Mohm**

Diagnosis information - Gateway control module

Battery sensor, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

000600

Diagnosis information - Gateway control module

Alternator, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

000601

Diagnosis information - Gateway control module

Alternator, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Gateway control module

Voltage converter, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Gateway control module

Voltage converter, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Gateway control module

Voltage converter, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

000607

Diagnosis information - Gateway control module

Voltage converter, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Gateway control module

Programming

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Incorrect data record
- ♦ Faulty data record

Fault setting condition(s)

The data record is checked after every programming operation.

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Gateway control module

Programming

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Incorrect data record
- ♦ Faulty data record

Fault setting condition(s)

The data record is checked after every programming operation.

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Gateway control module

Gateway control module, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

C11006

Diagnosis information - Gateway control module

Coding

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Coding incorrect or not carried out

Fault effects

Function not available or available only to a limited extent

C11008

Diagnosis information - Gateway control module

Gateway control module, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

C13000

Diagnosis information - Gateway control module

Current distributor, no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Power supply
- ♦ no communication

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

C13001

Diagnosis information - Gateway control module

Battery sensor, no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Power supply
- ♦ no communication

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Diagnosis information - Gateway control module

Alternator, no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ **Power supply**
- ♦ no communication

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Diagnosis information - Gateway control module

Voltage converter, no communication

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Power supply
- ♦ no communication

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

C13060

Diagnosis information - Gateway control module

Gateway control module, no communication LIN bus

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ no communication

Fault effects

Function not available or available only to a limited extent

Setpoint definitions

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Diagnosis information - Gateway control module

Current distributor, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

C13064

Diagnosis information - Gateway control module

Battery sensor, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Gateway control module

Alternator, Function restriction

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

Function not available or available only to a limited extent